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Regression Model Optimization using GridSearchCV

For this task, I have used the **Insurance Charges** dataset to build a regression model using Python and the Scikit-learn library. The objective of this task was to train a regression model and then improve its performance using hyperparameter tuning with **GridSearchCV**.

The dataset contains information about individuals such as age, gender, body mass index (BMI), number of children, smoking status, and region. These help to describe the characteristics of a person. The **target variable** in this dataset is **charges**, which represents the medical insurance cost.

Python libraries like **Pandas** and **NumPy** were used for data handling and calculations, **Scikit-learn** libraries such as **train_test_split**, **OneHotEncoder**, **RandomForestRegressor**, **GridSearchCV** and evaluation metrics were used to build, optimize, and evaluate the regression model.

Basic analysis steps were performed such as checking the first and last rows of the dataset for consistency check, viewing data types, shape and summary of the dataset. The dataset was also checked for missing values and duplicate rows. Duplicate records were removed.

Categorical columns, such as **sex**, **smoker**, and **region** were converted into numerical form. Since machine learning models require numerical data. These columns converted to numerical form using **One Hot Encoding**.

After preprocessing the data, the dataset was divided into two parts: **training data** and **testing data**. The training data was used to train the machine learning model, while the testing data was used to evaluate the performance.

A **Random Forest Regressor** was used as the baseline model. It is a machine learning method that uses many decision trees to make predictions. The model was trained with default settings and then used to predict values for the test data.

The performance of the model was evaluated using regression evaluation methods such as **R² score**, **Mean Absolute Error (MAE)**, and **Root Mean Squared Error (RMSE)**. The baseline model achieved an R² score of around **0.88**, which indicates that the model was able to explain most of the variation in the target variable.

To improve the model performance, **GridSearchCV** was used to tune the model parameters. **GridSearchCV** checks different combinations of parameters and finds the combination that gives the best results. Parameters like the number of trees, maximum depth of the trees, minimum samples required to split a node, and maximum number of features were tested.

After the grid search process was completed, the best model found by GridSearchCV was used to make predictions on the test dataset. The optimized model achieved an **R² score** of around **0.89**, which is slightly better than the baseline model.

This shows how machine learning models can be **improved using hyperparameter tuning**. By applying **GridSearchCV** to the **Random Forest model**, the **prediction performance improved slightly**.